# Bluetooth Audio Adaptor — Design Overview

**Rev B (in design)** · Updated 2026-08-30 · Jason Describes the **target design**. Deltas from as-built Rev A are listed in §10.

> Living document. Every confirmed design change lands here. Planning, open questions and review findings live in 0 and 1 — not here.

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## 1. Function

Bluetooth audio adaptor for a car AUX input. Powered from USB-C or an internal Li-ion cell.

| Mode | Path | Status |
|---|---|---|
| **RX** (primary) | Phone → Bluetooth A2DP → codec DAC → 3.5 mm out | Rev A hardware complete |
| **TX** | 3.5 mm in → codec ADC → Bluetooth → headphones/speaker | Rev B addition |
| **HFP** handsfree | On-board mic → codec ADC → BT; BT → DAC → jack | Rev B addition |

**Key constraint:** must work with all phones, not just recent ones → **Bluetooth Classic (A2DP/HFP), not BLE Audio**. This single requirement determines the MCU (§5).

**Out of scope:** amplified speaker output, multipoint pairing, app control.

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## 2. Architecture

**Power tree**

```
                    ┌──────────────────┐
                    │  Li-ion cell     │──── NTC ──► TS   temperature qualification
                    │  + 10 k NTC      │
                    └────────▲─────────┘
                             │ BAT ──[100k]──┬── BAT_ADC ──► ADC1
                             │               └──[100k]── GND
USB-C ── VBUS 5 V ──► ┌──────┴───────┐
  ├─ USBLC6 (ESD)     │   BQ24074    │  power path · DPPM
  ├─ CC1/CC2 ──► ADC1 │   charger    │  SYS clamped ≤ 4.4 V
  └─ C47 10 µF        └──┬───────┬───┘
                         │       ├── EN1 / EN2  ◄── GPIO   input-limit select
     ILIM ISET ──────────┘       ├── /PGOOD, /CHG ──► GPIO
     ITERM TMR                   │
                             SYS │ 3.0 – 4.4 V
                   ┌─────────┴─────────┐
                   ▼                   ▼
            ┌─────────────┐     ┌─────────────┐
            │   LDO A     │     │   LDO B     │
            │  AP7361C    │     │  LP5907     │
            │  SOT-89 1 A │     │  low noise  │
            └──────┬──────┘     └──────┬──────┘
                   │ 3V3_A             │ 3V3_B
                   │                   │
       ESP32 · flash · CH340K       ES8388 · MEMS mic
            · LEDs  (280 mA pk)          (20 mA)
```

### Signal chain

```
40 MHz XTAL ──┐        ┌── W25Q32JV 4 MB flash
              │        │
PCB antenna   │        │ SPI          USB-C ── USBLC6 ── CH340K
+ C-L-C match─┤        │                                    │
              ▼        ▼                                    │ UART
           ┌──────────────────────────┐                     │ + DTR/RTS
           │     ESP32-D0WD-V3        │◄────────────────────┘
           │  BT Classic + BLE        │
           └──────┬────────────┬──────┘
              I2C │        I2S │  (MCLK/SCLK/LRCK/DSDIN/ASDOUT)
                  ▼            ▼
           ┌──────────────────────────┐
           │   ES8388 stereo codec    │
           └───▲──────────┬───────▲───┘
     LIN1/RIN1 │  LOUT1/  │       │ LIN2/RIN2
               │  ROUT1   ▼       │
               │      ┌───────────┴───────┐
               │      │  TS5A23159 SPDT   │  TX / RX select
               │      └─────────┬─────────┘
               │                │ 1 µF AC coupling
               │                ▼
       on-board MEMS mic   3.5 mm TRS jack
       (differential pair)
```

### Rails

| Rail | Source | Loads | Why separate |
|---|---|---|---|
| SYS 3.0–4.4 V | BQ24074 | Both LDOs | Power path: USB when present, battery when not |
| 3V3_A | LDO A | ESP32, flash, CH340K, LEDs | Carries RF current bursts |
| 3V3_B | LDO B (LP5907) | ES8388, mic | Isolates a 96 dB codec from those bursts |

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## 3. Power subsystem

Linear regulation throughout. A switching converter was rejected: this is a mixed-signal audio board and the efficiency gain does not pay for the noise.

| Part | Function | Key spec |
|---|---|---|
| **BQ24074RGTR** (C54313) | Power-path charger | 1.5 A, SYS clamped ≤4.4 V, DPPM, NTC input, 10.5 V OVP |
| **AP7361C-33Y5-13** (C460397) | LDO A — digital/RF | SOT-89-5, 1 A, ultra-low dropout |
| **LP5907MFX-3.3** (C80670) | LDO B — codec analog | 6.5 µV<sub>RMS</sub>, high PSRR, no bypass cap needed |

| Part | Function | Key spec |
|---|---|---|
| **USBLC6-2SC6** | USB ESD | In-line on D+/D−; its `VBUS` pin stays at the connector, upstream of the charger |
| Li-ion cell + 10 k NTC | Energy | Removable. NTC bonded to cell, not PCB |

**VBUS reaches exactly three things:** the ESD array, the charger's `IN` pin, and `C47`. Every other load in the system hangs off `SYS`, including both LDOs. That is what makes the board behave identically on USB and on battery.

### 3.1 Power path — the five loops

In priority order. The system load always outranks the battery.

| Loop | Trips at | Behaviour |
|---|---|---|
| Input current limit | `EN1` / `EN2` + `R_ILIM` | Hard ceiling on total current into `IN`. Everything else lives inside this budget |
| VIN-DPM | V<sub>IN</sub> falls to 4.5 V | Reduces input current so a weak source cannot be crashed. **Active only in USB100/USB500 modes** |
| DPPM | V<sub>SYS</sub> falls to V<sub>O(REG)</sub> − 100 mV ≈ 4.3 V | Cuts *charge* current to hold SYS up. Charge termination is disabled while active |
| Battery supplement | V<sub>SYS</sub> < V<sub>BAT</sub> − 40 mV | Charge current already zero and the load still exceeds the input limit — BATFET turns fully on and the cell supplements. Exits above V<sub>BAT</sub> − 20 mV |
| Thermal regulation | T<sub>J</sub> ≥ 125 °C | Folds back charge current regardless of the above |

The last three slow the safety timers **in proportion to the charge-current reduction**. A deliberately low `R_ISET` gets no such slowdown — the timer would run at full speed while the cell trickled. This is why `R_ISET` is set *above* anything the input can deliver, and DPPM is left to mediate (§3.3).

### 3.2 Input current limit — mode strapping

`EN2` / `EN1` select the limit. Datasheet Table 7-2 orders the columns EN2, EN1:

| EN2 | EN1 | Limit | When |
|---|---|---|---|
| 0 | **1** | **475 mA (USB500)** | **Power-on default** — and any unknown or default-USB source |
| 1 | 0 | 1.073 A ( `R_ILIM` ) | Firmware has confirmed a ≥1.5 A source (§3.4) |
| 0 | 0 | 95 mA (USB100) | Strict compliance on an un-enumerated host port |
| 1 | 1 | — | Standby / USB suspend |

**The default must be USB500, not USB100.** Both pins floating gives USB100 through the chip's 285 kΩ internal pulldowns, and 100 mA cannot run a 171 mA system: with a flat cell the board would brown out before firmware ran and could never raise the limit. USB500 boots the system, matches what a PC port allows after enumeration, and is a mode that has VIN-DPM behind it.

```
SYS ──[R30 100k]──┬── EN1 ──► GPIO (drive open-drain)     default HIGH (3.26 V)
                  |
           (285k internal pulldown)

GND ──[R31 100k]──┴── EN2 ──► GPIO (push-pull)            default LOW
```

**100 kΩ, not 0 Ω** — the GPIO must be able to override the resistor. Against the internal 285 kΩ, EN1 sits at 4.4 × 285/385 = 3.26 V, comfortably over the 1.4 V V<sub>IH</sub>.

**Pull up to `SYS`, not `3V3_A`** — `3V3_A` does not exist until the system has booted, and the whole point of the default is to be defined before that. `SYS` is present whenever either source is, and at 4.4 V max sits inside the EN pin's 1.4–6 V window (7 V absolute max).

Drive `EN1` open-drain: the pull-up is to 4.4 V and the GPIO drives 3.3 V. Leave both pins alone whenever `/PGOOD` is high-Z — the EN inputs are ignored in power-down mode, and driving EN1 low would only burn cell current through R30.

### 3.3 Charge programming

| Resistor | Value | Formula | Result |
|---|---|---|---|
| `R_ISET` | 1.2 kΩ **1 %** | I<sub>CHG</sub> = K<sub>ISET</sub>/R, K = 890 | **742 mA** (664–812 over spread) |
| `R_ILIM` | 1.5 kΩ | I<sub>IN(max)</sub> = K<sub>ILIM</sub>/R, K = 1610 | **1.073 A** |
| `R_ITERM` | 3.0 kΩ | I<sub>TERM</sub> = 0.03 × R<sub>ITERM</sub>/R<sub>ISET</sub> | **75 mA** (≈C/27 on a 2000 mAh cell) |
| `R_TMR` | 56 kΩ | t<sub>MAXCHG</sub> = 10 × K<sub>TMR</sub> × R, K = 48 s/kΩ | **7.5 h** (5.6 h worst case) |

Pre-charge falls out of the same ISET resistor: K<sub>PRECHG</sub>/R<sub>ISET</sub> = 88/1200 = **73 mA**, the conventional 10 % of fast charge.

742 mA is 0.37C on a 2000 mAh cell and 0.25C on a 3000 mAh one — safe for either, so this resistor does not depend on the cell decision. Use 1 % on `R_ISET`: TI calls this out specifically to avoid tripping the internal RISET short-circuit test.

**`ILIM` must never be left open — an open `ILIM` pin disables all charging.** Worth an explicit bring-up check.

`R_NTC_LIN` is fitted **DNP** by design: it stands in for the pack NTC if a cell without one is used, or can be replaced by a high-value linearising resistor. It must not be populated alongside a real NTC — see §3.6.

### 3.4 Source capability detection

The default is USB500, so detection only ever has to answer one question: **is this a ≥1.5 A charger?** Everything else falls back safely.

**CC sensing (C-to-C only).** `R36`/`R37` tap CC1/CC2 through 1 kΩ into ADC1. With Rd = 5.1 kΩ:

| Source Rp | V<sub>CC</sub> | Advertised |
|---|---|---|
| 56 kΩ | 0.42 V | Default USB |

| Source Rp | V<sub>CC</sub> | Advertised |
|---|---|---|
| 22 kΩ | 0.94 V | 1.5 A |
| 10 kΩ | 1.69 V | 3.0 A |

Thresholds: >0.2 V attached, >0.66 V ≥ 1.5 A, >1.23 V = 3.0 A. Read **both** pins — only one carries the Rp, the other is open or is VCONN, and the pair also gives plug orientation. **No filter capacitor on the CC nodes:** the spec caps sink CC capacitance at ~200 pF. Multisample in software instead. Re-read periodically; a source may change its advertisement at any time.

**The limitation that shapes the architecture:** a USB-A-to-C cable is *required* by the spec to contain a 56 kΩ Rp, so CC always reports Default USB no matter how large the charger behind it. For the expected majority case — an A-to-C cable — CC sensing tells you nothing.

**BC1.2 (not fitted).** The only mechanism that unlocks a USB-A wall charger, by detecting that D+ is shorted to D− through ≤200 Ω (a Dedicated Charging Port). Needs a detector IC — BQ24392 or MAX14636 — whose integrated D+/D− isolation switch also solves the problem that the CH340K sits on that bus. Deferred; the EN1/EN2 GPIO control above is what keeps it a firmware-plus-one-IC change rather than a respin.

**Guard rail.** ILIM mode has no VIN-DPM, so a wrong detection has no graceful foldback. A VBUS divider into ADC1 lets firmware revert to USB500 if VBUS sags below ~4.5 V. **Do not** probe adaptively by stepping up and watching what happens — without VIN-DPM the failure mode is a brownout.

### 3.5 Safety timers

`R_TMR` = 56 kΩ gives a 7.5 h fast-charge timer (5.6 h worst case) and a 45 min pre-charge timer. Leaving `TMR` floating selects the internal default of 5 h typ / 4 h min; grounding it disables the timers entirely.

The timer must outlast the slowest legitimate charge. Worst case is a 3000 mAh cell charging in USB500 mode while playing: ~304 mA of charge current against 742 mA programmed is a 0.41 reduction, so the 5.6 h timer counts at 0.41× and covers 13.7 h of wall clock against an 8.9 h charge.

On expiry the part latches a fault and **`/CHG` flashes at ~2 Hz**. Firmware should decode that — it is also how a TS fault surfaces, and it is otherwise invisible.

### 3.6 Battery temperature qualification

Mandatory in hardware, not firmware. `TS` sources 75 µA into the pack thermistor and compares:

```
V_HOT  = 300 mV   →  R_TS ≤ 4.0 kΩ   →  too hot    (≈50 °C on a 10 k 103AT-2)
V_COLD = 2100 mV  →  R_TS ≥ 28 kΩ    →  too cold   (≈ 0 °C)
```

The valid window is therefore **4.0 kΩ – 28 kΩ**, which a bare 10 k Type-2 NTC maps onto 0–50 °C with 3 °C hysteresis. Any added network shifts those trip points: a 10 kΩ resistor in parallel with the NTC moves the hot trip to ~33 °C, and a 1 kΩ resistor pins R<sub>TS</sub> below 1 kΩ permanently, inhibiting all charging. **Fit the pack NTC or `R_NTC_LIN`, never both.**

### 3.7 Status and state of charge

`/PGOOD` and `/CHG` are open-drain, pulled up to **`3V3_A`** (not `SYS`) so both nets are clean 3.3 V logic a GPIO can read directly — pulling them to the 4.4 V SYS rail would exceed the ESP32's input rating.

| /PGOOD | /CHG | State |
|---|---|---|
| Hi-Z | Hi-Z | Running on battery |
| Low | Low | Charging |
| Low | Hi-Z | Charge complete, or suspended |
| Low | **2 Hz flash** | Fault — safety timer expired, or TS out of range |

**State of charge** is voltage-based: `R34` / `R35` form a 100 k / 100 k divider from `BAT` to ground, tapped as `BAT_ADC` (4.2 V → 2.1 V, inside ADC1's usable 150–2450 mV window at 11 dB attenuation). It costs 21 μA continuously, ~15 mAh/month — negligible against the cell. **100 nF at the tap** is required: 50 kΩ source impedance is high for the ESP32 SAR ADC. Calibrate with `esp_adc_cal` and the eFuse Vref, and multisample; raw ADC error is ±6 %, which is ±250 mV of cell voltage and useless.

Two limits define what this can deliver. The reading is meaningless while charging — the charger is driving V<sub>BAT</sub> toward 4.2 V — so firmware uses the `/PGOOD` + `/CHG` state instead and snaps to 100 % on termination. And the Li-ion OCV curve is flat from roughly 20 % to 80 % SoC, so voltage alone gives ±10–15 % there. **This is a four-segment gauge and a dependable low-battery warning, not a trustworthy percentage.** If a real percentage is ever needed over AVRCP, add a MAX17048 to the existing I$^2$C bus.

### 3.8 Power-related GPIO map

The budget is essentially full. Allocate deliberately.

| Signal | Dir | Pin class |
|---|---|---|
| `EN1`, `EN2` | out | any GPIO (`GPIO16` / `GPIO17`; avoid strapping pins) |
| `AUDIO_VCC_EN` | out | any GPIO. Pulled low by `R33`, so the codec rail is **off at boot** |
| CH340K gate | out | any GPIO — gate on a GPIO rather than `/PGOOD`, so firmware can isolate the bridge during BC1.2 detection |
| `/PGOOD`, `/CHG` | in | any input pin |
| `BAT_ADC`, `USB_CC1`, `USB_CC2`, VBUS sense | in | **ADC1 only** — ADC2 is unusable while the radio is active |

Four output-capable pins remain free (`GPIO4`, `16`, `17`, `21`) against exactly four outputs needed. Input-only ADC1 pins `GPIO34`, `36`, `39`, plus `GPIO32` / `33`, cover the inputs. There is no slack.

**Safety.** Charging is inhibited outside the cell's temperature window in hardware via the NTC on `TS` — not in firmware. Cell is removable; device is not stored in the vehicle.

**Why not LFP:** 3.2 V nominal sits below the 3.3 V rail for most of its discharge, forcing a boost converter in front of the codec.

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## 4. Audio subsystem

**ES8388** stereo codec — 24-bit, 96 dB DAC dynamic range, −83 dB THD+N. Chosen over the ESP32's internal DAC for quality, and over a DAC-only part because HFP needs an ADC.

| Signal | Connection |
|---|---|
| Control | I$^2$C, address **0x20** ( `CE` pulled to DGND) |
| Audio | I$^2$S, codec as **slave**; full duplex for HFP |
| Clocks | MCLK, SCLK, LRCK from ESP32 |
| Output | LOUT1/ROUT1 → 1 μF → 10 Ω → analog switch → jack |
| Input (TX) | Jack → switch → LIN2/RIN2 |
| Input (mic) | Differential LIN1/RIN1 |

**All four codec rails run at 3.3 V.** DVDD/PVDD at 1.8 V would put the codec's I/O in a different logic domain from the 3.3 V ESP32 — see §7.

| Part | Function | Key spec |
|---|---|---|
| **ES8388** | Stereo codec | QFN-28, 0.4 mm; AVDD/HPVDD/DVDD/PVDD all 3V3 |
| **SPH8878LR5H-1** (C3171733) | MEMS microphone | LGA-6, differential **or** single-ended, MSL 1 |
| **TS5A23159DGSR** (C42751) | TX/RX path switch | Dual SPDT, 1 Ω R<sub>on</sub>, break-before-make |
| 3.5 mm TRS jack | Audio I/O | 3-pole |

**Output coupling.** 1 μF blocks the VMID bias. Into a ~10 kΩ AUX load that is a ~16 Hz corner. Use a case size large enough that Class-II dielectric distortion stays below the codec's own THD.

**Ground loop.** Chassis ground (USB charger) and head-unit audio ground form a loop → alternator whine. Mitigations: run on battery with USB unplugged (free), or fit 1:1 isolation transformers in place of R7/R8. Both provisioned; footprints accept either.

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## 5. Microcontroller & RF

**ESP32-D0WD-V3.** QFN-48, 0.35 mm pitch, dual-core Xtensa LX6.

**Selected for firmware capability, not silicon features:** it is the only Espressif part whose radio hardware supports **Bluetooth Classic**, and the only one whose SDK provides an **A2DP sink** and **HFP**. ESP32-S3/C3/C6/H2 and the entire Nordic nRF range are BLE-only. This choice follows directly from §1's compatibility requirement.

| Part | Function | Key spec |
|---|---|---|
| **ESP32-D0WD-V3** | MCU + radio | BT Classic + BLE + Wi-Fi; needs external flash |
| **W25Q32JV** | Program store | 4 MB, 3.3 V ( `JV` not `JW` ), on VDD_SDIO |
| 40 MHz crystal | Reference | ±10 ppm required; load caps per §7 |
| PCB inverted-F antenna + C-L-C match | 2.4 GHz | 50 Ω controlled impedance, 4-layer |

**RF front end is a deliberate learning exercise.** A pre-certified module would remove the matching network, crystal trim and impedance routing — that is precisely the work being kept. Requires a NanoVNA (antenna S11) and a frequency-offset measurement to close.

**Stackup:** 4-layer. Signal / GND / PWR / Signal, with a solid ground plane under the RF section.

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## 6. Programming & debug

No USB peripheral on this ESP32 variant, and no SWD (Xtensa, not ARM).

| Path | Parts | Use |
|---|---|---|
| USB auto-program | CH340K + UMH3N dual transistor on DTR/RTS | Day-to-day flashing |
| Off-board serial | J3 header (DNP) | Fallback |
| JTAG | J4 2×5 header (DNP), ESP-PROG | Debugging |

Headers are DNP — fit before a debug session. 13 test points cover $I^2C$, mic, audio out, boot, GPIO and rails.

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## 7. Key calculations

### Power budget

| Load | Rail | Current |
|---|---|---|
| ESP32, BT Classic streaming (avg) | A | 130 mA |
| ESP32 transmit peak | A | 250 mA |
| SPI flash | A | 5 mA |
| CH340K (USB attached only) | A | 10 mA |
| 2× LED @ 330 Ω | A | 8 mA |
| **Rail A** | | **155 mA avg · 280 mA peak** |
| ES8388 @ 3.3 V, playback + record | B | 18 mA |
| MEMS mic | B | 0.4 mA |
| **Rail B** | | **~20 mA** |

**LDO A thermal.** Bluetooth bursts last 1–2 ms; package time constant is seconds, so the junction integrates to the average, not the peak.

```
P_avg = (4.4 − 3.3) × 0.155 = 0.17 W
SOT-89, θJA ≈ 100 °C/W  →  ΔT = 17 °C  →  Tj ≈ 87 °C at 70 °C ambient     ✓
SOT-23-5, θJA ≈ 200 °C/W →  ΔT = 34 °C  →  Tj ≈ 104 °C                     ✓ but tight
```

**LDO A dropout → usable battery.** Dropout, not current, is the binding spec.

```
Cutoff = 3.3 V + V_dropout(280 mA)

  150 mV → 3.45 V → ~87 % of cell
  250 mV → 3.55 V → ~83 %
  400 mV → 3.70 V → ~75 %
  1.2 V  → 4.50 V → will not run on battery       (NCP1117, Rev A — removed)
```

**Runtime** (LDO path, ~83 % usable): 1000 mAh ≈ 4.8 h · 2000 mAh ≈ 9.7 h · 3000 mAh ≈ 14.5 h.

**Charge current and time.** `R_ISET` programs 742 mA, but the input limit is what actually binds in every mode except ILIM. CC delivers ~85 % of capacity; the CV tail to a 75 mA termination adds ~0.5 h.

| Mode | Input | System | Charge | 2000 mAh | 3000 mAh |
|---|---|---|---|---|---|
| USB500, idle | 475 mA | ~20 mA | 455 mA | 4.2 h | 6.1 h |
| USB500, playing | 475 mA | 171 mA | 304 mA | 6.1 h | 8.9 h |
| ILIM, either | 1.073 A | 20–171 mA | **742 mA** (ISET binds) | 2.8 h | 3.9 h |

**Charger thermal.** Linear, so the charger burns (V_IN − V_BAT) × I_CHG plus the pass-FET loss. Worst case is the start of fast charge, V_BAT ≈ 3.0 V, in ILIM mode:

```
P = (5.0 − 4.4) × 0.913  +  (4.4 − 3.0) × 0.742  =  0.55 + 1.04  =  1.59 W
VQFN-16 RGT, θJA = 44.5 °C/W  →  ΔT = 71 °C  →  Tj ≈ 96 °C at 25 °C ambient    ✓
                                                Tj ≈ 116 °C at 45 °C ambient    ✓ (reg. point 12
USB500 mode: P = 0.71 W  →  ΔT = 32 °C  →  Tj ≈ 57 °C                          ✓
```

The input current limit is what guarantees this — raising `R_ISET` cannot overheat the part, because `ILIM` caps the current before it gets there.

**Crystal load capacitance.**

```
CL = (C1 × C2)/(C1 + C2) + C_stray       C_stray ≈ 2–5 pF

Rev A: 18 pF ∥ 18 pF → 9 + 3 ≈ 12 pF against a 10 pF crystal → over-loaded, frequency low
Rev B: ≈ 15 pF ∥ 15 pF → 7.5 + 3 ≈ 10.5 pF, then trim on measured offset (±10 ppm required)
```

**Logic-level compatibility (why all codec rails are 3.3 V).**

```
ESP32   VIH(min) = 0.75 × 3.3 = 2.475 V     VIL(max) = 0.825 V
Codec at PVDD 1.8 V: VOH ≈ 1.8 V  →  lands between VIL and VIH = undefined
Codec abs-max input = DVDD + 0.3 V = 2.1 V  →  3.3 V drive exceeds it
```

**Audio output coupling.** `f = 1/(2π · 10 kΩ · 1 µF) ≈ 16 Hz`.

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## 8. Conceptual circuits

**Microphone — differential, no external bias.** MEMS outputs are internally biased (0.66/0.70 V, 340/410 Ω). An electret-style bias resistor fights the internal amplifier and costs headroom.

```
3V3_B ──[100 Ω]──┬────────── MIC VDD         RC filter: mic PSRR is only −45 dB
                 │
              [1 µF]
                 │
                GND

OUT+ ──┬──┤├ 100 nF ──► ES8388 LIN1     Route OUT+/OUT− as a matched parallel
       │                                 pair — common-mode rejection depends on
    [10 nF]                              both picking up identical noise.
       │
      GND                                10 nF + 340 Ω ≈ 47 kHz — RF shunt only,
OUT− ──┬──┤├ 100 nF ──► ES8388 RIN1      above the audio band.
       │
    [10 nF]
       │
      GND
```

**TX/RX switch — placement is critical.**

```
            ┌──────────────┐
LOUT1 ──────┤ NO           │
            │   TS5A23159  ├── COM ──┤├ 1 µF ──► jack tip
LIN2  ──────┤ NC           │
            └──────┬───────┘
                   │ IN ── ESP32 GPIO
```

Switch sits on the **codec side** of the coupling capacitor. Both throws are at VMID (~1.65 V), inside the switch's 0–3.3 V supply rails. On the jack side the signal is centred at 0 V and every negative half-cycle would clip.

**RF match — C-L-C Pi, per Espressif.**

```
ESP32 LNA_IN ──┬──[ L series ]──┬────── PCB antenna
               │                 │
           [C shunt]         [C shunt]       Values tuned on measured S11.
               │                 │           50 Ω controlled-impedance trace.
              GND               GND
```

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## 9. Firmware dependencies

Hardware choices that constrain firmware, or vice versa.

| Item | Note |
|---|---|
| **Bluetooth Classic** | Only the original ESP32 has Classic radio hardware. A2DP sink + HFP come from ESP-IDF/ESP-ADF. Drives the entire MCU selection. |
| **Pairing persistence** | Bluedroid writes bonding keys to **NVS in flash** automatically. Survives power-off with no battery. Requires an initialised NVS partition. |
| **MCLK pin** | ESP32 can emit MCLK only on GPIO0/1/3. GPIO1/3 are the UART, so **GPIO0** — which is also the BOOT strapping pin. Safe: the codec's MCLK input is high-Z at reset. |
| **ADC** | Use **ADC1 only** (battery sense). ADC2 is unusable while the radio is active. GPIO32–39 free. |
| **Flash mode** | Configure **DIO**. QIO requires IO2/IO3 correct — fixed in Rev B, but DIO is the safe default. |
| **JTAG + GPIO12** | MTDI is the VDD_SDIO strapping pin. A debugger driving TDI high across reset selects 1.8 V and the 3.3 V flash fails. Fix: burn `XPD_SDIO_*` eFuses to force 3.3 V. **Irreversible.** |
| **Codec I²C address** | 0x20 (`CE` low). ESP-ADF's ES8388 driver default. |
| **I²S full duplex** | Required for HFP. Configure both TX and RX channels. |
| **TX/RX switch** | One GPIO selects the audio path. Firmware must track mode. |
| **Deep sleep** | If used for battery standby, populate the CAP2 RC network. |
| **Charger limit sequencing** | Boot default is USB500 (§3.2). Firmware may raise to ILIM mode only after confirming a ≥1.5 A source. Drive `EN1` open-drain; leave both pins alone when `/PGOOD` is high-Z. |
| **CC read** | `USB_CC1` / `USB_CC2` on **ADC1**. Read both, take the one in a valid band. An A-to-C cable always reports Default USB — do not treat that as a fault. |
| `/CHG` **fault** | A ~2 Hz flash is a latched fault (timer expiry or TS out of range), not "charging". Decode it; clear by toggling `CE` or the input. |
| **SoC blindness** | `BAT_ADC` is meaningless while charging. Report charger state instead, and snap to 100 % when `/CHG` releases. |
| **Codec rail** | `AUDIO_VCC_EN` is pulled low, so LDO B is **off at reset**. Firmware must enable it before touching the codec on I²C/I²S, or the ESP32 will back-power the ES8388 through its ESD diodes. |
| **CH340K gate** | Bridge is on a GPIO-controlled switch. Keep it off when USB is absent (~10 mA of battery budget) and during any BC1.2 detection. |

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## 10. Deltas from Rev A (as built)

| # | Change | Reason |
|---|---|---|
| 1 | Codec rails 1.8 V → 3.3 V; delete 1.8 V LDO, add dedicated codec LDO | Logic-level incompatibility + supply noise |
| 2 | `CE` pulled to DGND | Was floating — indeterminate I²C address |
| 3 | Add 10 nF on ESP32 `CAP1` | Required by datasheet; was absent |
| 4 | Swap flash IO2/IO3 | Crossed; corrupts QIO reads |
| 5 | Delete mic bias network (R21/R22/R23/C43/C44); add RC on mic VDD | Electret circuit on a self-biased part |
| 6 | Mic → SPH8878LR5H-1 | ICS-40720 discontinued |

| # | Change | Reason |
|---|---|---|
| 7 | LED resistors 1 kΩ → 330 Ω | ~0.3 mA was invisible |
| 8 | TRRS → 3-pole TRS jack | 4th pole unused |
| 9 | Add TX input path + analog switch | TX mode now in scope |
| 10 | Add battery charger, NTC, cell connector, VBAT sense | Battery in scope |
| 11 | NCP1117 → low-dropout LDO in SOT-89 | 1.2 V dropout cannot run from a cell |
| 12 | Crystal load caps 18 pF → ~15 pF, then trim | Over-loaded; ±10 ppm required |
| 13 | Charger input limit made firmware-selectable (`EN1`/`EN2` on GPIOs, 100 kΩ pulls) | Fixed strap cannot adapt to source capability; default USB500 boots on a flat cell |
| 14 | `/PGOOD`, `/CHG` pull-ups moved SYS → 3V3_A and routed to GPIOs | 4.4 V exceeds the ESP32 input rating; status was LED-only |
| 15 | Add `BAT_ADC` divider, CC1/CC2 sense taps | State of charge (REQ-PWR-10) and source detection |
| 16 | Gate the CH340K on a GPIO | ~10 mA of battery budget with USB absent |

**Still open:** cell choice (2000 mAh pouch vs 3000 mAh 18650) and therefore charge time; whether to fit BC1.2 detection; ground-loop mitigation (transformers vs battery-only playback); crystal offset measurement; RF match tuning; BOM line consolidation.

**Not yet on the schematic** (§3 describes the target): `R_ITERM` and `R_TMR` are still marked DNP and `R_TMR` still carries a 0 Ω value; the `BAT_ADC` filter cap and the VBUS sense divider are absent; the `EN1`/`EN2`/`/PGOOD`/`/CHG`/`BAT_ADC`/`USB_CC*` global labels have no counterparts on the micro sheet yet; and `+3V3_SYS` is still a separate net from the micro sheet's `+3V3`.

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## 11. Manufacturing

**JLCPCB**, 4-layer, **Economic PCBA** with the ESP32 excluded from assembly and hand-populated (it is the only Standard-Only part). All other parts verified Economic-tier.

Constraints: single-sided placement (satisfied — nothing on the back is assembled), 0402 minimum passive, 0.4 mm minimum IC pitch.

Build: 5 assembled + 5 bare. Target $20 USD landed per unit.

Hand assembly: 0.35 mm-pitch QFN requires a laser-cut stencil, plugged/capped via-in-pad on the thermal pad (open vias wick solder and degrade the RF ground), and ~50–70 % paste coverage on the exposed pad.

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## 12. References

**Local** (`hw/datasheets/`, `hw/app notes/`)


- ESP32 datasheet · ESP32 Hardware Design Guidelines · ESP32 Technical Reference Manual
- ES8388 datasheet rev 5.0 · ES8388 User Guide
- W25Q32JV datasheet rev G
- ESP32-LyraT v4.3 schematic — reference design for the codec section

- AN043 2.4 GHz PCB Antenna · Antennas for IoT

**External**

- JLCPCB PCB assembly capabilities
- ESP-ADF ES8388 driver
- esptool — set flash voltage eFuses

**Project**

- 0 — requirements, constraints, traceability
- 0 — netlist audit and findings
- 0 — battery architecture
- 0 — mic theory and selection

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## 13. Revision history

| Rev | Date | Change |
|---|---|---|
| A | 2025-04-05 | As-built schematic, commit `369c5f9` |
| B | 2026-08-24 | This document created. Scope decisions resolved; 12 deltas in §10 |
| B | 2026-08-30 | §3 expanded to cover the full charger configuration: power-path loops, EN1/EN2 strapping, charge programming, source detection, safety timers, TS qualification, SoC and GPIO map. Deltas 13–16 |

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